

SOEN 220 – OSI Model & TCP/IP Model

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1. Why Network Models?

Network models provide a structured way to understand how data travels from one device to another across a network. They divide the complex process of communication into smaller, manageable layers. Each layer has a specific function and interacts only with the layers directly above and below it. The two most important models are the **OSI Model** (a theoretical reference model) and the **TCP/IP Model** (the practical model that powers the Internet).

2. The OSI Model (Open Systems Interconnection)

Definition: The OSI Model is a conceptual framework developed by the International Organization for Standardization (ISO). It standardises the functions of a telecommunication or computing system into seven abstraction layers. It is primarily a **reference model** used for teaching and understanding network communication; it is not implemented as a single protocol suite.

The Seven Layers of the OSI Model (from top to bottom)

- 7. Application Layer** — Closest to the end user. Provides network services directly to applications (e.g. web browsers, email clients). Protocols: HTTP, FTP, SMTP, DNS, Telnet.
- 6. Presentation Layer** — Responsible for data translation, encryption/decryption and compression. Ensures that data from the application layer of one system can be read by the application layer of another. Formats data (ASCII, JPEG, etc.).
- 5. Session Layer** — Manages sessions (dialogues) between applications. Establishes, maintains and terminates connections. Handles synchronisation and checkpointing.
- 4. Transport Layer** — Provides end-to-end reliable (or unreliable) data transfer. Segments data, handles error recovery and flow control. Protocols: TCP (reliable), UDP (unreliable/fast).
- 3. Network Layer** — Responsible for logical addressing (IP addresses) and routing of packets across different networks. Determines the best path. Protocol: IP (IPv4/IPv6), ICMP, routers operate here.
- 2. Data Link Layer** — Provides node-to-node delivery. Frames data, handles physical addressing (MAC addresses), error detection, and media access control. Switches operate here. Sub-layers: LLC and MAC.
- 1. Physical Layer** — Deals with the actual transmission of raw bits over a physical medium (cables, radio waves, fibre). Defines voltages, pin layouts, cable specifications, and signalling.

Visual Stack – OSI 7 Layers

7. Application HTTP, FTP, SMTP, DNS
6. Presentation Encryption, compression, format
5. Session Session management & sync
4. Transport TCP / UDP – end-to-end delivery
3. Network IP addressing & routing
2. Data Link MAC, framing, switches
1. Physical Cables, bits, signals

Figure: OSI 7-layer reference model (top = closest to user)

3. The TCP/IP Model (Internet Protocol Suite)

Definition: The TCP/IP Model (also called the Internet Protocol Suite) is the practical set of communication protocols used on the Internet and most modern networks. It was developed by the U.S. Department of Defense and is named after its two core protocols: **Transmission Control Protocol (TCP)** and **Internet Protocol (IP)**. Unlike OSI, TCP/IP is a **working implementation** that actually runs on devices today.

The Four Layers of the TCP/IP Model

- 4. Application Layer** — Combines the functions of OSI's Application, Presentation and Session layers. Provides services to user applications. Protocols: HTTP, HTTPS, FTP, SMTP, DNS, SSH, DHCP, etc.
- 3. Transport Layer** — Same core role as OSI Transport. Provides process-to-process delivery. TCP offers reliable, connection-oriented service with flow control and error recovery. UDP offers fast, connectionless service.
- 2. Internet Layer (Network Layer)** — Handles logical addressing and routing of packets across networks. The key protocol is IP (IPv4 and IPv6). Also includes ICMP (diagnostics) and IGMP. Routers operate mainly at this layer.
- 1. Network Access / Host-to-Network Layer** — Combines OSI's Data Link and Physical layers. Deals with the physical transmission of data over the network medium and framing. Protocols/standards: Ethernet, Wi-Fi (802.11), PPP, ARP.

Visual Stack – TCP/IP 4 Layers

4. Application HTTP, FTP, SMTP, DNS, SSH
3. Transport TCP (reliable) / UDP (fast)
2. Internet IP addressing & routing
1. Network Access Ethernet, Wi-Fi, physical media

Figure: TCP/IP 4-layer model (the model actually used on the Internet)

4. OSI vs TCP/IP – Side-by-Side Comparison

OSI Layer	TCP/IP Layer	Key Protocols / Devices
7. Application 6. Presentation 5. Session	Application	HTTP, FTP, SMTP, DNS, SSH
4. Transport	Transport	TCP, UDP
3. Network	Internet	IP, ICMP, routers
2. Data Link 1. Physical	Network Access	Ethernet, Wi-Fi, switches, cables

Table: How OSI layers map onto the TCP/IP model

Key Differences

- Purpose:** OSI is a theoretical reference model for understanding and standardising communication. TCP/IP is a practical protocol suite that is actually implemented and used worldwide.
- Number of layers:** OSI has 7 layers; TCP/IP has 4 layers (some texts show 5 by splitting Network Access into Data Link + Physical).
- Development:** OSI was developed by ISO as a standard. TCP/IP was developed by DARPA and became the de-facto standard of the Internet.
- Reliability focus:** OSI places more emphasis on reliability at several layers. TCP/IP concentrates reliability mainly in the Transport layer (TCP).
- Usage today:** Almost all modern networks and the Internet run on TCP/IP. OSI is mainly used for teaching, documentation and as a conceptual guide.

5. Data Encapsulation (How Data Travels)

When a user sends data (e.g. an email or web request), the data is passed down the layers. Each layer adds its own header (and sometimes a trailer). This process is called **encapsulation**. At the receiving end the process is reversed (**de-encapsulation**) and each layer removes its header.

- Application data** → Transport segment (TCP/UDP header added) → Network packet (IP header added) → Data-link frame (MAC header + trailer) → Physical bits on the wire/wireless medium.
- Remember the PDU names: **Data** (Application) → **Segment** (Transport) → **Packet** (Network/Internet) → **Frame** (Data Link) → **Bits** (Physical).

6. Quick Revision Checklist

- I can name all 7 OSI layers in the correct order (Application → Physical).
- I can name the 4 TCP/IP layers and map them to the corresponding OSI layers.
- I know that OSI is a reference model while TCP/IP is the practical Internet suite.

- I can state the main function of each OSI layer in one sentence.
- I know the key protocols that operate at the Application, Transport and Internet layers.
- I understand the difference between TCP (reliable, connection-oriented) and UDP (unreliable, connectionless).
- I can explain encapsulation and the names of the PDUs at each layer (Data → Segment → Packet → Frame → Bits).
- I can list at least three practical differences between the OSI and TCP/IP models.